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Título/Title Program ANCOLAB, version 4.x User's manual	
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Resumen/Summary The ANCOLAB program (ANalysis of COMposite materials: LAMinates Basic) is a computer procedure for the analysis of laminates made of composite materials. This version of the manual, corresponding to version 4 of the program, introduces the following modifications: • New PC version • Only one laminate in each run • New and simpler input data file • NASTRAN cards only in PC version This report describes how to use the ANCOLAB program version 4.x, and includes examples to further clarify its use. PROPIEDAD DE EADS CASA. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Proyectos y Sistemas de EADS CASA. EADS CASA PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorisation by EADS CASA Engineering and System Direction.

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(*).- Cover sheet only	Aprobado Approved	Darío Fernández Villalba

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1 Technical sheet of the program

Technical Report NT-T-ADP-01031

PROGRAMREFERENCEDATE

ANCOLAB

March 1993

OBJECTIVE

The ANCOLAB program is a computer procedure for the analysis of laminates made of composite materials.

The program offers the following types of analysis of the laminates:

- Average properties of the laminate
- Stiffness matrices
- Stress-Strain analysis
- Failure analysis (first layer or progressive)

LANGUAGECOMPUTER

Fortran 77 and Visual Basic

Sun Solaris, PC Windows

NUMBER OF LINES (APROX.)EXECUTION MODE

8,000 lines

Interactive

AUTHORVERSION

F. Lecina
J.C. Gómez

4.0

ADDITIONAL DOCUMENTS

Technical Report: C/CM/86-028 B:
Manual del Programador del Programa ANCOLAB
Version 2.1

Technical Report: C/CM/86-029 B:
Manual Teórico del Programa ANCOLAB
Version 2.1

2 Introduction

This manual describes the use of version 4 of the ANCOLAB program, including examples to further clarify its use. The program calculates, in all cases, the average properties and stiffness matrices of the laminate. If the input data file includes load cases the program will do Stress-Strain or Loads-Moments analysis. If it is included one or more failure criteria in the load cases, the program will do the strength analysis based on first-layer failure or based on progressive failure (only for tapes). The program can also calculate the MAT2, MAT8 and PCOMP cards for NASTRAN.

The figure below illustrates the system of axes and the notation used.

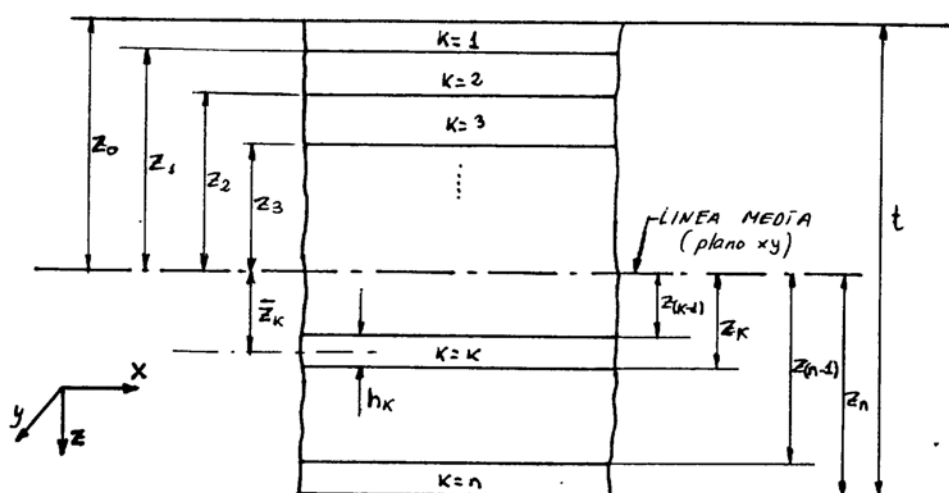


Figure 2.1.1 Notation of the program

Note: The vertical positions of the layers are measured from the X-Y plane through the mean line of the laminate (marked "LINEA MEDIA (plano xy)" in the diagram)

It should be noted that the sign convention adopted for the moments is the same as that used by the PANPA, COMPASION and ARSA programs, and is the opposite of that used by NASTRAN.

However, for the layer codes, the NASTRAN criterion has been used. This is the opposite of the criteria used by PANPA, COMPASION and ARSA.

Currently, there is a version of the program installed on the Multivac Computer and another on the PC Server of EADS CASA's DPS.

The main program was developed in Fortran 77. However, the version of the program installed on the PC Server has a more "friendly" user interface for data entry and results output; this graphical interface was developed in Visual Basic. Therefore, the calculation modules of both versions are exactly the same.

3 Program execution

3.1 PC Server version

The Visual Basic executable program (**ancolab.exe**) is located in:

<\\gpr60005\InfoCalculo\Soft\Ancolab>

In the folder \\gpr60005\InfoCalculo\Soft\Ancolab\Consola, there is another executable program, which is the Fortran program.

This latter program is started by the first one, so to execute the program, the user has to click twice on the Visual Basic executable program.

To ease this process, it is suggested creating a direct access to the program. To do so, drag with the mouse (do not copy) the **ancolab.exe** program file to a directory (the Desktop, for example). In this case, to execute the program, the user just has to click twice on this shortcut.

3.2 Multivac Computer version

In order to execute the Multivac Program, it is necessary to previously assign the calculation symbols. The following steps must be followed:

1. Enter the following lines on the **.profile** file:

```
ENV=$HOME/.kshrc
export ENV
BASH_ENV=$HOME/.bashrc
export BASH_ENV
. asignaciones_calculo
```

2. If the user works with the **ksh** shell, the following line must be entered in the **.kshrc** file:

```
. asignaciones_calculo
```

3. If the user works with the **bash** shell, the following line must be entered in the **.bashrc** file:

```
. asignaciones_calculo
```

4. Enter the following command:

```
. .profile
```

(Or exit and enter the MULTIVAC System again)

Once this is made, the program is executed by means of the following command:

```
$ancolab
```

4 Data input

4.1 PC Server version

In the PC version, when the program starts, the following information window is displayed:

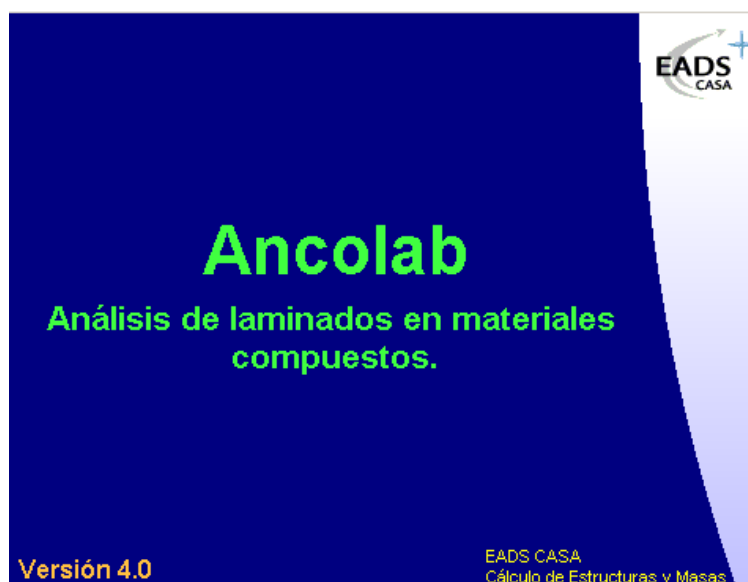


Figure 4.1.1 Program start

The input data can be entered via a file or using the graphical interface. If an input file is used, it is necessary to press “Open” button from the Menu of the program and to choose the name of the file. If the graphical interface is used, first the “Material Properties” button must be pressed and a form appears like the one shown below. Then, the boxes corresponding to each material property must be filled. If a strength analysis is wanted, the material allowables have to be different from 0.

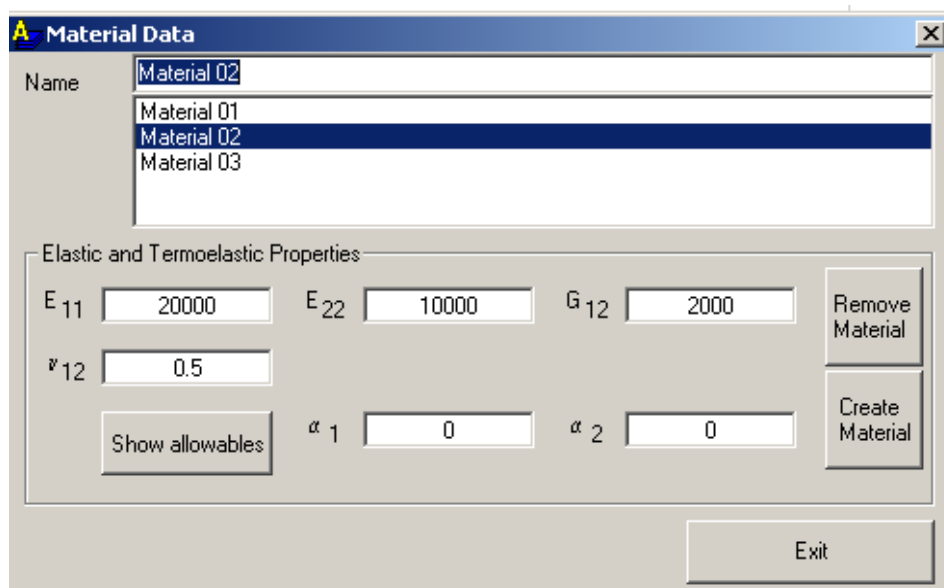
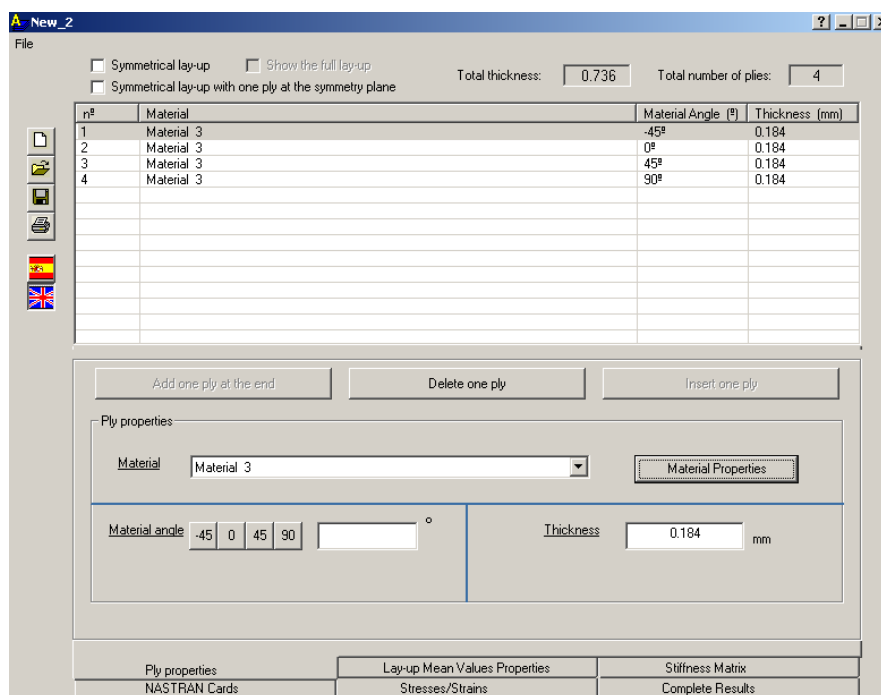


Figure 4.1.2 Material data form

When all the boxes are filled, the user has to press “Create Material”. Next, the button “Exit” might be pressed to finish the material input data entry or the boxes might be again filled to create a new material.

After the thickness and the angle of the ply are chosen, press the button “Add ply at the end”. This way the user can create a laminate using the buttons and the check that appear on the screen. Here there is an example of a laminate:



n°	Material	Material Angle (°)	Thickness (mm)
1	Material 3	-45°	0.184
2	Material 3	0°	0.184
3	Material 3	45°	0.184
4	Material 3	90°	0.184

Total thickness: 0.736 Total number of plies: 4

Buttons: Add one ply at the end, Delete one ply, Insert one ply

Ply properties:

Material: Material 3 Material Properties

Material angle: -45 0 45 90 ° Thickness: 0.184 mm

Bottom tabs: Ply properties (selected), Lay-up Mean Values Properties, Stiffness Matrix, NASTRAN Cards, Stresses/Strains, Complete Results

Figure 4.1.3 Ply properties form

The user can only press the rest of the tabs when the laminate is completed.

4.2 Multivac Computer version

Data are entered via a data file.

Every time the program is run, it creates a data file with a ".DAT" extension that contains the data used in the analysis, properly ordered. This file can be used as the data source to repeat the same analysis, or may be used as a basis for a new data file, to perform a new analysis.

When the user starts the Multivac computer version of the program, this appear on the screen:

```
Program ANCOLAB
=====
```

Procedure to analyse laminates made of composite materials.

In the laminate analysis, the program presents the following options:

- MEAN VALUES OF THE PROPERTIES OF THE LAMINATE
- STIFFNESS MATRIX
- STRESSES - STRAINS
- FAILURE ANALYSIS (FIRST PLY OR PROGRESSIVE)

Version 4,0 September-2005

Notes

- This software and the data produced by it are property of EADS-CASA. The total or partial reproduction of this software is forbidden as well as its use out of the EADS-CASA environment.

The input data file must be introduced interactively via the keyboard:

Name of the input data file	→
-----------------------------	---

The full name of the file containing the output data must be given.

Name of the output file without extension [ANCOLAB] →

The program generates 2 output files:

".RES": A file containing the results of the analysis.

".DAT": A file containing the data used for the analysis.

If the user simply presses <RETURN>, the program uses the default filename "ANCOLAB".

4.3 Data input file

The input data file has the following characteristics:

- Everything to the right of a "!" character is assumed to be a comment. The user can add or omit as many comments as he desired.
- Data are entered in free format (separated by commas or blank).
- The continuation symbol is "+". When this character is placed at the end of a line, it indicates that the data on that line continues on the following line.
- The measurement units used must be coherent.

The input file can be divided in three groups of data: materials, laminate definition and loads. The data of each group are shown below:

	Title
M A T E R I A L S	N_m
	$E_{11}(1), E_{22}(1), E_{12}(1), \mu(1), \alpha_1(1), \alpha_2(1)$
	$L-t_e(1), L-c_e(1), T-t_e(1), T-c_e(1), Sh_e(1), L-t_d(1), L-c_d(1), T-t_d(1), T-c_d(1), Sh_d(1)$
	.
	.
	$E_{11}(N_m), E_{22}(N_m), E_{12}(N_m), \mu(N_m), \alpha_1(N_m), \alpha_2(N_m)$
	$L-t_e(N_m), L-c_e(N_m), T-t_e(N_m), T-c_e(N_m), Sh_e(N_m), L-t_d(N_m), L-c_d(N_m), T-t_d(N_m), T-c_d(N_m), Sh_d(N_m)$
L A M I N A T E	N_l
	Material code
	Angle code
	Thickness code
L O A D S	N_{lc}
	or $\left\{ \begin{array}{l} 1, N_x(1), N_y(1), N_{xy}(1), M_x(1), M_y(1), M_{xy}(1), \Delta T(1), \text{failure criterion}(1) \\ 2, D_{fx}(1), D_{fy}(1), D_{fxy}(1), K_{px}(1), K_{py}(1), K_{pxy}(1), \Delta T(1), \text{failure criterion}(1) \end{array} \right.$
	.
	.
	or $\left\{ \begin{array}{l} 1, N_x(N_{lc}), N_y(N_{lc}), N_{xy}(N_{lc}), M_x(N_{lc}), M_y(N_{lc}), M_{xy}(N_{lc}), \Delta T(N_{lc}), \text{failure criterion}(N_{lc}) \\ 2, D_{fx}(N_{lc}), D_{fy}(N_{lc}), D_{fxy}(N_{lc}), K_{px}(N_{lc}), K_{py}(N_{lc}), K_{pxy}(N_{lc}), \Delta T(N_{lc}), \text{failure criterion}(N_{lc}) \end{array} \right.$
	.
	$L-t_c, L-c_c, T-t_c, T-c_c, Sh_c$

Next, each line of the file is explained:

Title.- It is the identifier of the laminate. The title given here will appear at the head of each page of the results output file (maximum 78 characters).

MATERIALS

N_m.- It is the number of materials used in the laminate. A maximum of 8 different materials can be used in the laminate.

Materials average properties and allowables.- These data must be entered on two lines for each of the different materials defined above, in free format (separated by commas or blanks). The first line contains the following average properties:

- E_{11} : Elastic modulus in the longitudinal direction.
- E_{22} : Elastic modulus in the transversal direction.
- E_{12} : Shear modulus.
- μ : Poisson's ratio
- α_1 : Thermal expansion coefficient in the longitudinal direction.
- α_2 : Thermal expansion coefficient in the transversal direction.

The second line contains the material allowable values. These allowables are defined for each material as stresses and strains. The values used in the analysis will depend on the chosen failure criterion (see point 9): strains for failure types 1, 2, 6 or 7 and stresses for failure types 3, 4 or 5. If these values are not necessary for the calculation, they can be zero, but in all cases, values must be given for all these data items.

Stresses	Strains
$L-t_e, L-c_e, T-t_e, T-c_e, Sh_e$	$L-t_d, L-c_d, T-t_d, T-c_d, Sh_d$
$L-t_e$: Allowable tensile stress in the longitudinal direction.	
$L-c_e$: Allowable compression stress in the longitudinal direction.	
$T-t_e$: Allowable tensile stress in the transversal direction.	
$T-c_e$: Allowable compression stress in the transversal direction.	
Sh_e : Allowable shear stress.	
$L-t_d$: Allowable tensile strain in the longitudinal direction.	
$L-c_d$: Allowable compression strain in the longitudinal direction.	
$T-t_d$: Allowable tensile strain in the transversal direction.	
$T-c_d$: Allowable compression strain in the transversal direction.	
Sh_d : Allowable shear strain.	

LAMINATE

N_l.- It's the number of layers of the laminate. Enter the total number of layers or plies that constitute the laminate. A maximum of 200 layers is admitted.

Material code.- It is the number corresponding to each material of the laminate corresponding to N.E.L. notation. A maximum of 600 characters is admitted.

E.g.: 10 layers of 5 different materials: $((1/2/3)(2)/(4/5)(2)) = (1/2/3/1/2/3/4/5/4/5)$
 16 layers of the same material: $(1)(16)$

Angle code.- It is, in N.E.L. notation, the angle formed between the axis system associated with each layer and the coordinate system of the laminate. A maximum of 600 characters is admitted.

E.g. 10 layers: $(45/-45/90/0/90)S = (45/-45/90/0/90/90/0/90/-45/45)$

Thickness code.- It is the nominal thickness of each layer of the laminate in N.E.L. notation. A maximum of 600 characters is admitted.

E.g. 10 layers: $((0.1/0.2/0.3)\$)S = (0.1/0.2/0.3/0.2/0.1/0.1/0.2/0.3/0.2/0.1)$
 16 layers of the same thickness: $(0.18)(16)$

LOADS

N_{lc}. It is the number of load cases to be analysed. A maximum of 30 load cases is admitted. If the number of load cases is 0, the program only calculates average properties and stiffness matrices.

Load cases data.- One line must be entered for each load case, where:

- 1 or 2 is the type of external loads given: 1 (loads per unit length) or 2 (strains)
- $(N_x, N_y, N_{xy}, M_x, M_y, M_{xy})$ or $(Df_x, Df_y, Df_{xy}, Kp_x, Kp_y, Kp_{xy})$ are the values of the external loads.
- ΔT is the temperature increase value.
- The failure criterion is an integer and can be:
 - 0.- No strength check
 - 1.- Maximum strain modified
 - 2.- Maximum strain
 - 3.- Tsai-Hill
 - 4.- Tsai-Wu modified
 - 5.- Hoffman.
 - 6.- EADS CASA/Plates
 - 7.- Progressive failure

It can be used any combination of the criteria above, except for the criterion number 7.
 E.g.: 36 means that the strength of the laminate will be checked using criteria 3 and 6.

The progressive failure of the laminate (only applied in tape) uses the criterion of maximum strains in a layer and it is calculated using only the first load case. When a failure occurs in a layer, in direction 2 or 1-2, the stiffness values E_{22} and G_{12} are assumed to be zero, and a new iteration is performed, to determine the next failure in the matrix in the 1 or 1-2 direction. This process continues until the convergence or, if the process doesn't converge, until a failure in the direction of the fibres is found or the maximum number of iterations is exceeded.

If the failure criterion is 0 the program doesn't do strength analysis.

CASA/Plates allowable strains.- The allowable strains corresponding to failure criterion 6 (CASA/Plates) must be given, on one line and in free format. If this criterion is not used in any load case, any value may be entered, since the program will ignore this data. The values must be given in the following order:

L-t_c, L-c_c, T-t_c, T-c_c, Sh_c

- L-t_c: Allowable tensile strain in the longitudinal direction.
- L-c_c: Allowable compression strain in the longitudinal direction.
- T-t_c: Allowable tensile strain in the transversal direction.
- T-c_c: Allowable compression strain in the transversal direction.
- Sh_c: Allowable shear strain.

5 Results and output files

5.1 PC Server version

In the PC version, when the laminate is completed the rest of the tabs can be pressed and the different results appear. The following figures show an example of these outputs:

The screenshot shows the 'Lay-up Mean Values Properties' form. At the top, there are checkboxes for 'Symmetrical lay-up' and 'Show the full lay-up', and a checkbox for 'Symmetrical lay-up with one ply at the symmetry plane'. The 'Total thickness' is 0.736 and the 'Total number of plies' is 4. Below this is a table with 4 columns: 'nº', 'Material', 'Material Angle (°)', and 'Thickness (mm)'. The table contains 4 rows of data. Below the table is a section for 'Lay-up Mean Values Properties' with fields for E_{xx} , E_{yy} , G_{xy} , N_{xy} , N_{yx} , α_x , α_y , and α_{xy} . At the bottom, there are tabs for 'Ply properties', 'Lay-up Mean Values Properties', 'Stiffness Matrix', 'NASTRAN Cards', 'Stresses/Strains', and 'Complete Results'.

nº	Material	Material Angle (°)	Thickness (mm)
1	Material 3	-45°	0.184
2	Material 3	0°	0.184
3	Material 3	45°	0.184
4	Material 3	90°	0.184

Lay-up Mean Values Properties

E_{xx} 16666.3500 N_{xy} 0.5124888 α_x 0.000

E_{yy} 15528.3000 N_{yx} 0.4774938 α_y 0.000

G_{xy} 5379.2500 α_{xy} 0.000

Ply properties Lay-up Mean Values Properties Stiffness Matrix
NASTRAN Cards Stresses/Strains Complete Results

Figure 5.1.1 Lay-up mean values properties form

The screenshot shows the 'Stiffness Matrix' form. At the top, there are checkboxes for 'Symmetrical lay-up' and 'Show the full lay-up', and a checkbox for 'Symmetrical lay-up with one ply at the symmetry plane'. The 'Total thickness' is 0.736 and the 'Total number of plies' is 4. Below this is a table with 4 columns: 'nº', 'Material', 'Material Angle (°)', and 'Thickness (mm)'. The table contains 4 rows of data. Below the table is a section for 'Stiffness Matrix' with four matrices: Matrix A, Matrix B, Matrix C, and Matrix D. At the bottom, there are tabs for 'Ply properties', 'Lay-up Mean Values Properties', 'Stiffness Matrix', 'NASTRAN Cards', 'Stresses/Strains', and 'Complete Results'.

nº	Material	Material Angle (°)	Thickness (mm)
1	Material 3	-45	0.184
2	Material 3	0	0.184
3	Material 3	45	0.184
4	Material 3	90	0.184

Stiffness Matrix

Matrix A

16875.43	8358.856	0.00
8358.856	16875.43	0.00
0.00	0.00	4258.286

Matrix B

-391.7623	-188.6263	290.1942
-188.6263	769.015	290.1942
290.1942	290.1942	-188.6263

Matrix C

-391.7623	-188.6263	290.1942
-188.6263	769.015	290.1942
290.1942	290.1942	-188.6263

Matrix D

654.9879	377.3299	-53.39575
377.3299	868.5708	-53.39576
-53.39575	-53.39576	192.2247

Ply properties Lay-up Mean Values Properties Stiffness Matrix
NASTRAN Cards Stresses/Strains Complete Results

Figure 5.1.2 Stiffness Matrix form

The screenshot shows the 'NASTRAN Cards' form. At the top, there are checkboxes for 'Symmetrical lay-up' and 'Show the full lay-up', and a checkbox for 'Symmetrical lay-up with one ply at the symmetry plane'. The 'Total thickness' is 0.736 and the 'Total number of plies' is 4. Below this is a table with columns 'nº', 'Material', 'Material Angle', and 'Thickness'.

nº	Material	Material Angle	Thickness
1	Material 03	-45	0.184
2	Material 03	0	0.184
3	Material 03	45	0.184
4	Material 03	90	0.184

Below the table is a section for 'NASTRAN Cards' containing a text area with the following content:

```

$ 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 .. 8 .. 9 .. 10 ..
MAT2 11 22928.6 11357.1 0.0 22928.6 0.0 5785.7
MAT2 22 19714.3 11357.1 -1607.1 26142.9 -1607.1 5785.7
MAT2 33 723.2 348.2 -535.7 -1419.6 -535.7 348.2
$
$ 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 .. 8 .. 9 .. 10 ..
MAT8 1 10000 50000.500000 1000 A001001H
+001001H 0.0E+00 0.0E+00 0.0
MAT8 2 20000 100000.500000 2000 A001002H
+001002H 0.0E+00 0.0E+00 0.0
MAT8 3 30000 150000.500000 3000 A001003H
+001003H 0.0E+00 0.0E+00 0.0
$
  
```

At the bottom, there are tabs for 'NASTRAN Cards', 'Stresses/Strains', and 'Complete Results'. Below these are sub-tabs: 'Ply properties', 'Lay-up Mean Values Properties', and 'Stiffness Matrix'.

Figure 5.1.3 NASTRAN cards form

If the “Load Cases” tab is clicked appears a table like this:

The screenshot shows the 'Load Cases' form. At the top, there are checkboxes for 'Symmetrical lay-up' and 'Show the full lay-up', and a checkbox for 'Symmetrical lay-up with one ply at the symmetry plane'. The 'Total thickness' is 0.736 and the 'Total number of plies' is 4. Below this is a table with columns 'nº', 'Material', 'Material Angle (°)', and 'Thickness'.

nº	Material	Material Angle (°)	Thickness
1	Material 3	-45	0.184
2	Material 3	0	0.184
3	Material 3	45	0.184
4	Material 3	90	0.184

Below the table is a section for 'Load Cases' containing a table with columns 'Stresses' and 'Strains'.

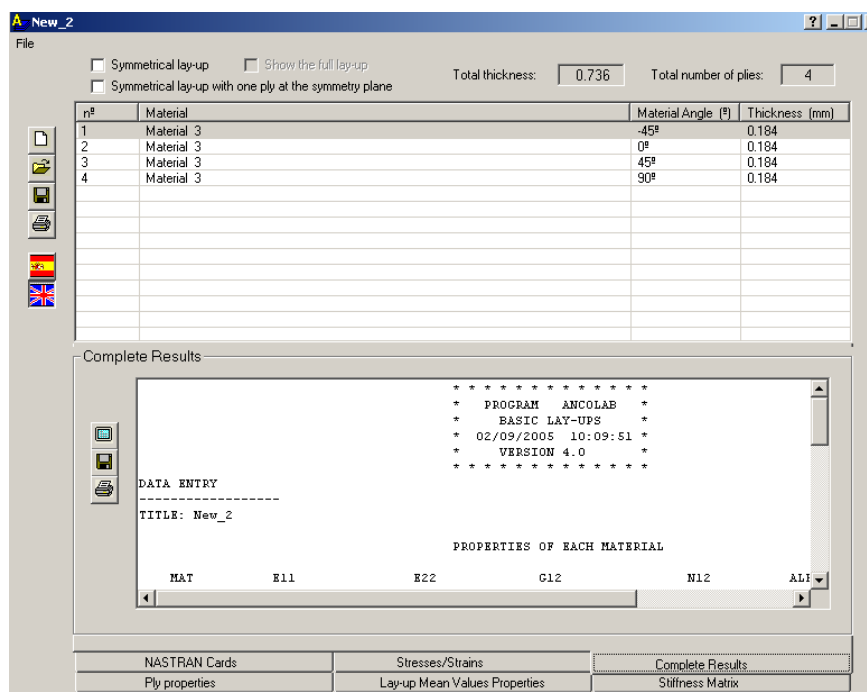
Stresses	Strains
Nx	Def x
Ny	Def y
Nxy	Def xy
Mx	Kx
My	Ky
Mxy	Kxy

Below the table is a 'Calculate Load Case' button. To the right of the table is a 'Failure Criteria' section with a 'First Ply Failure' checkbox and a list of failure criteria: 'Maximum strain (modified)', 'Maximum strain', 'Tsai - Hill', 'Tsai - Wu (modified)', 'Hoffman', 'CASA/Plates', and 'Progressive Failure'.

At the bottom, there are tabs for 'NASTRAN Cards', 'Stresses/Strains', and 'Complete Results'. Below these are sub-tabs: 'Ply properties', 'Lay-up Mean Values Properties', and 'Stiffness Matrix'.

Figure 5.1.4 Load cases form

First, the type of data must be chosen: strain or stresses. Depend on the option chosen one side of the table will be enabled. Next, the user has to enter the strains or stresses and the failure criteria of the load case number one. If no failure criterion is checked, the program doesn't do the strength analysis. All the results are shown pressing the tab "Completed Results". The user can print or save them.



The screenshot shows the 'New_2' software window. At the top, there are checkboxes for 'Symmetrical lay-up' and 'Show the full lay-up', and a checkbox for 'Symmetrical lay-up with one ply at the symmetry plane'. Below these are input fields for 'Total thickness: 0.736' and 'Total number of plies: 4'. A table lists materials with their angles and thicknesses:

nº	Material	Material Angle (º)	Thickness (mm)
1	Material 3	-45º	0.184
2	Material 3	0º	0.184
3	Material 3	45º	0.184
4	Material 3	90º	0.184

Below the table is the 'Complete Results' section, which contains a 'DATA ENTRY' area with a 'TITLE: New_2' and a 'PROPERTIES OF EACH MATERIAL' section with a table of material properties:

MAT	E11	E22	G12	N12	AL1

At the bottom of the window, there are tabs for 'NASTRAN Cards', 'Stresses/Strains', 'Complete Results', 'Ply properties', 'Lay-up Mean Values Properties', and 'Stiffness Matrix'.

Figure 5.1.5 Complete results form

5.2 Multivac Computer version

Every time the program is run, it generates two files, named:

<Name>.DAT and <Name>.RES

where <Name> is the filename that was entered when the program requested a name for the output files. The default filename is "ANCOLAB".

5.3 Output file

The content of the output files is the following:

- 1) File <Name>.DAT

This file contains the data used for the analysis, in the proper order. This file can be used as the data input file to repeat the same analysis, or it may be used as a model for a new data file, with suitable changes in the data. In the PC version, this file is created when the user selects “Save data” on the Menu of the program.

- 2) File <Name>.RES

This file contains the values of the data used for the analysis and the results produced by it.

The first part of the file contains the input data used by the program:

- The title of the laminate
- The average properties of each of the materials used in the laminate
- The structure of the laminate, giving the thickness and angle of each layer.

Next, the file contains the results of the analysis divided in three different parts: average properties and stiffness matrices, stress/strain or load/moments calculation and strength analysis. The first part is always in the file and the existence of the other two parts depends on the type of analysis that was performed. The explanation of each part of the file is the following:

- **Average properties and stiffness matrices;** the program calculates the average properties of the laminate based on its structure and on the properties of the material in each layer. The values obtained, referred to the global X-Y co-ordinates of the laminate, are:

E_{xx} : Elastic modulus of the laminate in the X direction

E_{yy} : Elastic modulus of the laminate in the Y direction

G_{xy} : Shear modulus of the laminate

N_{xy} : Poisson's major ratio

N_{yx} : Poisson's minor ratio

AlphaX: Coefficient of thermal expansion in the X direction.

AlphaY: Coefficient of thermal expansion in the Y direction.

AlphaXY: Coefficient of thermal expansion in the XY direction.

The program also calculates the laminate stiffness matrix which depends on the material, orientation, thickness and on the sequence of stacking of its layers. The stiffness matrix consists of three matrices, each 3 by 3 in sizes, which are known as the A, B and D matrices.

$$\begin{bmatrix} A & B \\ B & D \end{bmatrix}$$

where:

[A]= Stiffness matrix in the plane (membrane stiffness)

[B] = Membrane-Bending coupling matrix

[D] = Bending stiffness matrix

- **Stress-Strain or Loads-Moments;** these are only calculated by the program if there are one or more load cases defined in the input data file. In this case, the results are the loads, if strains are provided, or the strains, if loads are provided, based on the A, B and D stiffness matrices. The equations used are:

$$\begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \cdot \begin{bmatrix} D \\ K \end{bmatrix}$$

$$\begin{bmatrix} D \\ K \end{bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix}^{-1} \cdot \begin{bmatrix} N \\ M \end{bmatrix}$$

where:

N: N_x, N_y, N_{xy} (Forces per unit length)

M: M_x, M_y, M_{xy} (Moments per unit length)

D: Df_x, Df_y, Df_{xy} (Strains)

K: K_x, K_y, K_{xy} (Curvatures)

- **Strength analysis;** if there is defined a failure criterion different from 0 in one of the load cases, the program calculates the reserve factors of each layer, based on the real stresses or strains and the allowable stresses and strains. Several different failure criteria may be used for this analysis. If the first ply failure criterion is chosen, the failure is diagnosed when the reserve factor is less than 1 in at least one of the layers.

5.4 NASTRAN cards

The PC version of the program ANCOLAB can generate the three MAT2 cards that correspond to the laminate and the PCOMP card, together with the MAT8 cards corresponding to each of the layers in the laminate. These cards can be used as part of a BULK DATA DECK containing input data for NASTRAN. To obtain these cards the user has to press the tab “NASTRAN cards” after complete the data of the laminate. Then the cards appear and they can be saved in a file or printed. The names of the cards are 11, 22, 33 for MAT2 cards and 44 for the PCOMP card.

The user should refer to the NASTRAN User's Manual for a description of these cards.

Note that:

- The MAT2 cards are three cards that represents three materials whose material property matrices are the stiffness matrices of the laminate divided by a factor. A MAT2 card has the following format:

MAT2	MID	G11	G12	G13	G22	G23	G33	RHO
	A1	A2	A3	TREF	GE	ST	SC	SS

Only the terms of the matrix corresponding to the properties of the material are written to the MAT2 cards.

The first MAT2 card represents a material whose property matrix is formed by the terms of the stiffness matrix A of the laminate divided by the total thickness of the panel:

$$\begin{bmatrix} G_{11}^1 & G_{12}^1 & G_{13}^1 \\ & G_{22}^1 & G_{23}^1 \\ & & G_{33}^1 \end{bmatrix} = \frac{1}{t} \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ & A_{22} & A_{23} \\ & & A_{33} \end{bmatrix}$$

The second MAT2 card represents a material whose property matrix is formed by the terms of the stiffness matrix D of the laminate divided by the factor $t^3/12$ (where t = total thickness of the panel):

$$\begin{bmatrix} G_{11}^2 & G_{12}^2 & G_{13}^2 \\ & G_{22}^2 & G_{23}^2 \\ & & G_{33}^2 \end{bmatrix} = \frac{12}{t^3} \begin{bmatrix} D_{11} & D_{12} & D_{13} \\ & D_{22} & D_{23} \\ & & D_{33} \end{bmatrix}$$

The third MAT2 card represents a material whose property matrix is formed by the terms of the stiffness matrix B of the laminate divided by the factor t^2 (where t = total thickness of the panel):

$$\begin{bmatrix} G_{11}^3 & G_{12}^3 & G_{13}^3 \\ & G_{22}^3 & G_{23}^3 \\ & & G_{33}^3 \end{bmatrix} = \frac{1}{t^2} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ & B_{22} & B_{23} \\ & & B_{33} \end{bmatrix}$$

- Only the values for E_1 , E_2 , μ_{12} , G_{12} , α_1 and α_2 corresponding to the materials in each layer that form the laminate are written to the MAT8 cards.

- For the PCOMP card, it is assumed that:
 - a) The laminate is centred with respect to the reference plane.
 - b) The sequence of the layers (from bottom to top) coincides with the order in the input data given.
 - c) The system of axes for the material coincides with the reference system adopted for the ANCOLAB program.
 - d) Neither stress nor strength calculations are requested.
- In both the MAT8 cards and the PCOMP card the reference temperature is 0°.

Below it is given an example of NASTRAN cards from ANCOLAB:

NASTRAN CARDS

=====

```

$ 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 .. 8 .. 9 .. 10 .
MAT2 11 22928.6 11357.1 0.0 22928.6 0.0 5785.7
MAT2 22 19714.3 11357.1 -1607.1 26142.9 -1607.1 5785.7
MAT2 33 723.2 348.2 -535.7 -1419.6 -535.7 348.2
$
$ 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 .. 8 .. 9 .. 10 .
MAT8 1 10000 50000.500000 1000 A001001M
+001001M 0.0E+00 0.0E+00 0.0
MAT8 2 20000 100000.500000 2000 A001002M
+001002M 0.0E+00 0.0E+00 0.0
MAT8 3 30000 150000.500000 3000 A001003M
+001003M 0.0E+00 0.0E+00 0.0
$
$ 1 .. 2 .. 3 .. 4 .. 5 .. 6 .. 7 .. 8 .. 9 .. 10 .
PCOMP 44 0.0 A001001P
+001001P 3 0.184 -45.000 3 0.184 0.000 A001002P
+001002P 3 0.184 45.000 3 0.184 90.000
$
$ NOTES:
$ -----
$ 1. - Only the terms of the matrix corresponding to the properties
$ of the material are written to the MAT2 cards.
$
$ 2. - Only the values for E1, E2, NU12, G12, A1 y A2 corresponding to the
$ materials in each of the layers that form the laminate are written
$ to MAT8 card.
$
$ 3. - For the PCOMP card, it is assumed that:
$ a) The laminate is centred with respect to the reference plane.
$ b) The sequence of the layers (from bottom to top) coincides with the order
$ in the input data given to the ANCOLAB program.
$ c) The system of axes for the material coincides with the reference
$ system adopted for data entry into the ANCOLAB program.
$ d) Neither stress nor strength calculations are requested.
$
$ 4. - In both the MAT8 card and the PCOMP card the reference temperature is 0°.

```

6 Examples

Below there are 4 examples of the use of this program, one for each type of analysis. It is hoped that these examples will clarify the information given in this manual.

In the first example the program only calculates the average properties and the stiffness matrices because the load cases number is zero.

EXAMPLE 1

In this first example, average properties and stiffness matrix of this laminate are calculated:

(45/-45/0/90/0/90/-45/45)_s

Material: Group V Tape.

Data are given to the program by a file and the program generates other two files:

EJ_ANCOLAB_1.DAT, EJ_ANCOLAB_1.RES

The data file given to the program is:

File EJ_ANCOLAB_1.DAT

EJEMPLO 1

1

138000. 8900.00 4700.00 0.300 -1.3000E-06 3.1300E-05
0.0 0.0 0.0 0.0 0.0 0.000000 0.000000 0.000000 0.000000 0.000000

16

(1)(16)

(45/-45/0/90/0/90/-45/45)S

(.180)(16)

0

0 0.000E+00 0.000E+00 0.000E+00 0.000E+00 0.000E+00 0.000E+00 0.000E+00 0
0.000000 0.000000 0.000000 0.000000 0.000000

```

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!
!          <<ANCOLAB>> PROGRAM INPUT DATA FILE
!          -----
!
! Everything to the right of a "!" character is assumed to be a comment. !
! The user can add or omit as many comments as desired.                !
! Data is entered in free format.                                       !
!
! The continuation symbol is "+". When this character is placed at the !
! end of a line, it indicates that the data on that line continues on   !
! on the following line.                                                !
!
! The units used must be coherent.                                     !
!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!
! The identifier (title) of the laminate (maximum 78 characters)
EJEMPLO 1
!
! 2) The number of materials used in the laminate (maximum 16)
1
!
! For each material two lines have been entered:
!
!      E11, E22, E12, μ, ALFA1, ALFA2
!      L-t, L-c, T-t, T-c, Shear           L-t, L-c, T-t, T-c, Shear
!
! Where the first line must contain the following average properties
! of each material used in the laminate and second line must contain
! the allowable values of the materials used in the laminate:
!
!              Stresses                      Strains
!      -----
!      L-t, L-c, T-t, T-c, Sh             L-t, L-c, T-t, T-c, Sh
!
! L=Longitudinal, T=Transversal, t=tension, c=compression.
!
! If the failure criteria 1,2 or 7 are not chosen in any load case,
! any value may be entered for the allowable strains (the program
! does not use them).
! If the failure criteria 3,4 or 5 are not chosen in any load case,
! any value may be entered for the allowable stresses (the program
! does not use them).
! If strength analysis is not required any value may be entered for the
! allowables, because the program does not use them.
! In any case, values must be given for all these data items.
138000.    8900.00    4700.00    0.300    -1.3000E-06    3.1300E-05
  0.0     0.0     0.0     0.0     0.0 0.000000 0.000000 0.000000 0.000000 0.000000
!
! The number of layers of the laminate (maximum 200)
16
!
! The material associated with each layer of the laminate, corresponding to N.E.L. notation.
! Example: 10 layers of 5 different materials: ((1/2/3)(2)/(4/5)(2)) = (1/2/3/1/2/3/4/5/4/5)
! Maximum number of characters: 600.
(1)(16)
!
! The angle associated with each layer of the laminate, corresponding to N.E.L. notation.
! Example: 10 layers: (45/0/0/45/90)S = (45/0/0/45/90/90/45/0/0/45)
! Maximum number of characters: 600.

```

```

(45/-45/0/90/0/90/-45/45)S
!
! The thickness associated with each layer of the laminate, corresponding to N.E.L. notation.
! Example: 10 layers: ((0.1/0.2/0.3)$)S = (0.1/0.2/0.3/0.2/0.1/0.1/0.2/0.3/0.2/0.1)
! Maximum number of characters: 600.
(.180)(16)
!
! The number of load cases (maximum 30)
0
!
! For each different load case the following data must be entered:
!   a) Type of external load given :
!       1 (loads per unit length)
!       0
!       2 (strains)
!   b) The values of the external loads:
!       Nx, Ny, Nxy, Mx, My, Mxy
!       0
!       Dfx, Dfy, Dfxy, Kpx, Kpy, Kpxy
!   c) The temperature increase value
!   d) The failure criterion:
!       0.- No strength check
!       1.- Maximum strain (modified)
!       2.- Maximum strain
!       3.- Tsai-Hill
!       4.- Tsai-Wu (modified)
!       5.- Hoffman
!       6.- CASA/Plates
!       7.- Progressive failure
!       It can be used any combination of the above (except for the criterion
!       number 7).
!       Example: 36 means that the strength of the laminate will be
!               checked using criteria 3 and 6.
0  0.      0.      0.      0.      0.      0.      0.      0
!
! Allowable strains corresponding to the failure criterion number 6,
! in the following order:
!
!   L-t, L-c, T-t, T-c, Shear
!
! where: L=Longitudinal, T=Transversal, t=tension, c=compression.
! If the criterion number 6 is not chosen in any load cases, any
! value may be entered (they are not used by the program).
0.000000 0.000000 0.000000 0.000000 0.000000

```


File EJ_ANCOLAB_1.RES

```

* * * * *
*   PROGRAM   ANCOLAB   *
*   BASIC LAY-UPS      *
* 21- 09-2006 12:14:23*
*   VERSION 4.0        *
* * * * *

```

DATA ENTRY

TITULO: EJEMPLO 1

PROPERTIES OF EACH MATERIAL

MAT	E11	E22	G12	N12	ALFA1	ALFA2
1	138000.000	8900.000	4700.000	0.3000000	-1.300E-06	3.130E-05

LAY-UP COMPOSITION

PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS
1	1	45.00	0.180	2	1	-45.00	0.180	3	1	0.00	0.180	4	1	90.00	0.180
5	1	0.00	0.180	6	1	90.00	0.180	7	1	-45.00	0.180	8	1	45.00	0.180
9	1	45.00	0.180	10	1	-45.00	0.180	11	1	90.00	0.180	12	1	0.00	0.180
13	1	90.00	0.180	14	1	0.00	0.180	15	1	-45.00	0.180	16	1	45.00	0.180

MEAN VALUES PROPERTIES OF THE LAY-UP

Exx	Eyy	Gxy	Nxy	Nyx	Alfax	Alfay	Alfaxy
52802.652	52802.652	20148.303	0.3103498	0.3103498	1.178E-06	1.178E-06	7.359E-14

STIFFNESS MATRIX

MATRIX A				MATRIX B				MATRIX D			
168279.875	52225.625	0.000	0.000	0.000	0.000	122665.992	41864.426	4543.850			
52225.625	168279.891	0.000	0.000	0.000	0.000	41864.426	98432.148	4543.848			
0.000	0.000	58027.125	0.000	0.000	0.000	4543.850	4543.848	45874.426			

EXAMPLE 2

In the second example one load case is given, so the program calculates the strains/stresses produced by this load case in the laminate.

Below the data file and the .RES file are shown. The .DAT output file is similar to the file shown before changing the data.

File EJ_ANCOLAB_2.DAT

EJEMPL0 2

1

138000. 8900.00 4700.00 0.300 -1.3000E-06 3.1300E-05
0.0 0.0 0.0 0.0 0.0 0.000000 0.000000 0.000000 0.000000 0.000000

16

(1)(16)

(45/-45/0/90/0/90/-45/45)S

(.180)(16)

1

2 6.500E-03 4.300E-03 3.100E-03 0.000E+00 0.000E+00 0.000E+00 0.000E+00 0
0.000000 0.000000 0.000000 0.000000 0.000000

File EJ_ANCOLAB_2.RES

```

* * * * *
*   PROGRAM  ANCOLAB  *
*   BASIC LAY-UPS   *
* 21- 09-2006 12:19:52*
*   VERSION 4.0     *
* * * * *

```

DATA ENTRY

TITULO: EJEMPLO 2

PROPERTIES OF EACH MATERIAL

MAT	E11	E22	G12	N12	ALFA1	ALFA2
1	138000.000	8900.000	4700.000	0.3000000	-1.300E-06	3.130E-05

LAY-UP COMPOSITION

PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS
1	1	45.00	0.180	2	1	-45.00	0.180	3	1	0.00	0.180	4	1	90.00	0.180
5	1	0.00	0.180	6	1	90.00	0.180	7	1	-45.00	0.180	8	1	45.00	0.180
9	1	45.00	0.180	10	1	-45.00	0.180	11	1	90.00	0.180	12	1	0.00	0.180
13	1	90.00	0.180	14	1	0.00	0.180	15	1	-45.00	0.180	16	1	45.00	0.180

MEAN VALUES PROPERTIES OF THE LAY-UP

Exx	Eyy	Gxy	Nxy	Nyx	Alfax	Alfay	Alfaxy
52802.652	52802.652	20148.303	0.3103498	0.3103498	1.178E-06	1.178E-06	7.359E-14

STIFFNESS MATRIX

MATRIX A				MATRIX B				MATRIX D			
168279.875	52225.625	0.000	0.000	0.000	0.000	122665.992	41864.426	4543.850			
52225.625	168279.891	0.000	0.000	0.000	0.000	41864.426	98432.148	4543.848			
0.000	0.000	58027.125	0.000	0.000	0.000	4543.850	4543.848	45874.426			

LOAD CASES

=====

Caso: 1

Nx = 1.318E+03	Mx = 0.00	Temp= 0.00
Ny = 1.063E+03	My = 0.00	
Nxy = 180.	Mxy = 0.00	
Defx = 6.500E-03	Kpx = 0.00	
Defy = 4.300E-03	Kpy = 0.00	
Defxy = 3.100E-03	Kpxy = 0.00	

EXAMPLE 3

In this example, safety margins of each layer, with three failure criteria, numbers 1, 2 and 5 (Maximum strain modified and Hoffman) are calculated.

Data are introduced by means of a file. File of example 1 is used with a modification about the analysis to perform. Commentary lines have been deleted of this file.

Next, are included the input data flow and the files generated by the program:

EJ_ANCOLAB_3.DAT, EJ_ANCOLAB_3.RES

File EJ_ANCOLAB_3.DAT

EJEMPLO_3

1

138000. 8900.00 4700.00 0.300 -1.3000E-06 3.1300E-05
1680 1135 55 0.0 85 0.012174 0.010245 0.006180 0.005000 0.018085

16

(1)(16)

(45/-45/0/90/0/90/-45/45)S

(.180)(16)

2

2 8.750E-03 6.300E-03 3.200E-03 0.000E+00 0.000E+00 0.000E+00 0.000E+00 12
2 3.400E-03 7.700E-03 4.500E-03 0.000E+00 0.000E+00 0.000E+00 0.000E+00 5
0.000000 0.000000 0.000000 0.000000 0.000000

File EJ_ANCOLAB_3.RES

```

* * * * *
*   PROGRAM  ANCOLAB  *
*   BASIC LAY-UPS   *
* 21- 09-2006 12:20:40*
*   VERSION 4.0      *
* * * * *

```

DATA ENTRY

TITULO: EJEMPLO_3

PROPERTIES OF EACH MATERIAL

MAT	E11	E22	G12	N12	ALFA1	ALFA2
1	138000.000	8900.000	4700.000	0.3000000	-1.300E-06	3.130E-05

LAY-UP COMPOSITION

PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS
1	1	45.00	0.180	2	1	-45.00	0.180	3	1	0.00	0.180	4	1	90.00	0.180
5	1	0.00	0.180	6	1	90.00	0.180	7	1	-45.00	0.180	8	1	45.00	0.180
9	1	45.00	0.180	10	1	-45.00	0.180	11	1	90.00	0.180	12	1	0.00	0.180
13	1	90.00	0.180	14	1	0.00	0.180	15	1	-45.00	0.180	16	1	45.00	0.180

MEAN VALUES PROPERTIES OF THE LAY-UP

Exx	Eyy	Gxy	Nxy	Nyx	Alfax	Alfay	Alfaxy
52802.652	52802.652	20148.303	0.3103498	0.3103498	1.178E-06	1.178E-06	7.359E-14

STIFFNESS MATRIX

MATRIX A				MATRIX B				MATRIX D			
168279.875	52225.625	0.000	0.000	0.000	0.000	122665.992	41864.426	4543.850			
52225.625	168279.891	0.000	0.000	0.000	0.000	41864.426	98432.148	4543.848			
0.000	0.000	58027.125	0.000	0.000	0.000	4543.850	4543.848	45874.426			

LOAD CASES

=====

Caso: 1

```

Nx = 1.801E+03      Mx = 0.00      Temp= 0.00
Ny = 1.517E+03      My = 0.00
Nxy = 186.          Mxy = 0.00
Defx = 8.750E-03     Kpx = 0.00
Defy = 6.300E-03     Kpy = 0.00
Defxy = 3.200E-03    Kpxy = 0.00

```

Caso: 2

```

Nx = 974.           Mx = 0.00      Temp= 0.00
Ny = 1.473E+03      My = 0.00
Nxy = 261.          Mxy = 0.00
Defx = 3.400E-03     Kpx = 0.00
Defy = 7.700E-03     Kpy = 0.00
Defxy = 4.500E-03    Kpxy = 0.00

```

LOAD CASE NUMBER 1
=====

MAXIMUM STRAIN CRITERION (MODIFIED)

PLY	AD/L-T	AD/L-C	AD/T-T	AD/T-C	AD/CORT	DEF-1	DEF-2	DEF-12	R.F.-1	R.F.-2	R.F.-12
----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
1	0.012174	0.010245	0.006180	0.050000	0.018085	0.009125	0.005925-0.002450		1.3341	1.0430	4.4874
2	0.012174	0.010245	0.006180	0.050000	0.018085	0.005925	0.009125 0.002450		2.0547	0.6773	4.4874
3	0.012174	0.010245	0.006180	0.050000	0.018085	0.008750	0.006300 0.003200		1.3913	0.9810	4.4874
4	0.012174	0.010245	0.006180	0.050000	0.018085	0.006300	0.008750-0.003200		1.9324	0.7063	4.4874
5	0.012174	0.010245	0.006180	0.050000	0.018085	0.008750	0.006300 0.003200		1.3913	0.9810	4.4874
6	0.012174	0.010245	0.006180	0.050000	0.018085	0.006300	0.008750-0.003200		1.9324	0.7063	4.4874
7	0.012174	0.010245	0.006180	0.050000	0.018085	0.005925	0.009125 0.002450		2.0547	0.6773	4.4874
8	0.012174	0.010245	0.006180	0.050000	0.018085	0.009125	0.005925-0.002450		1.3341	1.0430	4.4874
9	0.012174	0.010245	0.006180	0.050000	0.018085	0.009125	0.005925-0.002450		1.3341	1.0430	4.4874
10	0.012174	0.010245	0.006180	0.050000	0.018085	0.005925	0.009125 0.002450		2.0547	0.6773	4.4874
11	0.012174	0.010245	0.006180	0.050000	0.018085	0.006300	0.008750-0.003200		1.9324	0.7063	4.4874
12	0.012174	0.010245	0.006180	0.050000	0.018085	0.008750	0.006300 0.003200		1.3913	0.9810	4.4874
13	0.012174	0.010245	0.006180	0.050000	0.018085	0.006300	0.008750-0.003200		1.9324	0.7063	4.4874
14	0.012174	0.010245	0.006180	0.050000	0.018085	0.008750	0.006300 0.003200		1.3913	0.9810	4.4874
15	0.012174	0.010245	0.006180	0.050000	0.018085	0.005925	0.009125 0.002450		2.0547	0.6773	4.4874
16	0.012174	0.010245	0.006180	0.050000	0.018085	0.009125	0.005925-0.002450		1.3341	1.0430	4.4874

MAXIMUM STRAIN CRITERION

PLY	AD/L-T	AD/L-C	AD/T-T	AD/T-C	AD/CORT	DEF-1	DEF-2	DEF-12	R.F.-1	R.F.-2	R.F.-12
----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
1	0.012174	0.010245	0.006180	0.050000	0.018085	0.009125	0.005925-0.002450		1.3341	1.0430	7.3816
2	0.012174	0.010245	0.006180	0.050000	0.018085	0.005925	0.009125 0.002450		2.0547	0.6773	7.3816
3	0.012174	0.010245	0.006180	0.050000	0.018085	0.008750	0.006300 0.003200		1.3913	0.9810	5.6516
4	0.012174	0.010245	0.006180	0.050000	0.018085	0.006300	0.008750-0.003200		1.9324	0.7063	5.6516
5	0.012174	0.010245	0.006180	0.050000	0.018085	0.008750	0.006300 0.003200		1.3913	0.9810	5.6516
6	0.012174	0.010245	0.006180	0.050000	0.018085	0.006300	0.008750-0.003200		1.9324	0.7063	5.6516
7	0.012174	0.010245	0.006180	0.050000	0.018085	0.005925	0.009125 0.002450		2.0547	0.6773	7.3816
8	0.012174	0.010245	0.006180	0.050000	0.018085	0.009125	0.005925-0.002450		1.3341	1.0430	7.3816
9	0.012174	0.010245	0.006180	0.050000	0.018085	0.009125	0.005925-0.002450		1.3341	1.0430	7.3816
10	0.012174	0.010245	0.006180	0.050000	0.018085	0.005925	0.009125 0.002450		2.0547	0.6773	7.3816
11	0.012174	0.010245	0.006180	0.050000	0.018085	0.006300	0.008750-0.003200		1.9324	0.7063	5.6516
12	0.012174	0.010245	0.006180	0.050000	0.018085	0.008750	0.006300 0.003200		1.3913	0.9810	5.6516
13	0.012174	0.010245	0.006180	0.050000	0.018085	0.006300	0.008750-0.003200		1.9324	0.7063	5.6516
14	0.012174	0.010245	0.006180	0.050000	0.018085	0.008750	0.006300 0.003200		1.3913	0.9810	5.6516
15	0.012174	0.010245	0.006180	0.050000	0.018085	0.005925	0.009125 0.002450		2.0547	0.6773	7.3816
16	0.012174	0.010245	0.006180	0.050000	0.018085	0.009125	0.005925-0.002450		1.3341	1.0430	7.3816

LOAD CASE NUMBER 2
=====

HOFFMAN CRITERION

PLY	AD/L-T	AD/L-C	AD/T-T	AD/T-C	AD/CORT	ESF-1	ESF-2	ESF-12	R. F.
----	-----	-----	-----	-----	-----	-----	-----	-----	-----
1	1680.0	1135.0	55.0	0.0	85.0	1091.5	50.5	20.2	1.0893
2	1680.0	1135.0	55.0	0.0	85.0	479.0	78.7	-20.2	0.6990
3	1680.0	1135.0	55.0	0.0	85.0	492.6	78.1	21.1	0.7046
4	1680.0	1135.0	55.0	0.0	85.0	1077.9	51.1	-21.2	1.0760
5	1680.0	1135.0	55.0	0.0	85.0	492.6	78.1	21.1	0.7046
6	1680.0	1135.0	55.0	0.0	85.0	1077.9	51.1	-21.2	1.0760
7	1680.0	1135.0	55.0	0.0	85.0	479.0	78.7	-20.2	0.6990
8	1680.0	1135.0	55.0	0.0	85.0	1091.5	50.5	20.2	1.0893
9	1680.0	1135.0	55.0	0.0	85.0	1091.5	50.5	20.2	1.0893
10	1680.0	1135.0	55.0	0.0	85.0	479.0	78.7	-20.2	0.6990
11	1680.0	1135.0	55.0	0.0	85.0	1077.9	51.1	-21.2	1.0760
12	1680.0	1135.0	55.0	0.0	85.0	492.6	78.1	21.1	0.7046
13	1680.0	1135.0	55.0	0.0	85.0	1077.9	51.1	-21.2	1.0760
14	1680.0	1135.0	55.0	0.0	85.0	492.6	78.1	21.1	0.7046
15	1680.0	1135.0	55.0	0.0	85.0	479.0	78.7	-20.2	0.6990
16	1680.0	1135.0	55.0	0.0	85.0	1091.5	50.5	20.2	1.0893

EXAMPLE 4

In this example safety margins of each layer are calculated, considering the criterion of failure progressive. This option is only available to laminates with only tapes.

Data are introduced by means of a file. File of example 1 is used with a modification about the load cases and the failure criterion. Commentary lines have been deleted of this file.

Next, are included the input data flow and the files generated by the program:

EJ_ANCOLAB_4.DAT, EJ_ANCOLAB_4.RES

File EJ_ANCOLAB_4.DAT

EJEMPLO 4

1

138000. 8900.00 4700.00 0.300 -1.3000E-06 3.1300E-05
0.0 0.0 0.0 0.0 0.0 0.012174 0.010245 0.006180 0.005000 0.018085

16

(1)(16)

(45/-45/0/90/0/90/-45/45)S

(.180)(16)

1

2 7.400E-03 6.700E-03 4.500E-03 0.000E+00 0.000E+00 0.000E+00 0.000E+00 7
0.000000 0.000000 0.000000 0.000000 0.000000

File EJ_ANCOLAB_4.RES

```

* * * * *
*   PROGRAM  ANCOLAB  *
*   BASIC LAY-UPS    *
* 21- 09-2006 12:50:33*
*   VERSION 4.0      *
* * * * *

```

ITERATION NUMBER: 1

DATA ENTRY

TITULO: EJEMPLO 4

PROPERTIES OF EACH MATERIAL

MAT	E11	E22	G12	N12	ALFA1	ALFA2
-----	-----	-----	-----	-----	-----	-----
1	138000.000	8900.000	4700.000	0.3000000	-1.300E-06	3.130E-05

LAY-UP COMPOSITION

PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
1	1	45.00	0.180	2	1	-45.00	0.180	3	1	0.00	0.180	4	1	90.00	0.180
5	1	0.00	0.180	6	1	90.00	0.180	7	1	-45.00	0.180	8	1	45.00	0.180
9	1	45.00	0.180	10	1	-45.00	0.180	11	1	90.00	0.180	12	1	0.00	0.180
13	1	90.00	0.180	14	1	0.00	0.180	15	1	-45.00	0.180	16	1	45.00	0.180

MEAN VALUES PROPERTIES OF THE LAY-UP

Exx	Eyy	Gxy	Nxy	Nyx	Alfax	Alfay	Alfaxy
-----	-----	-----	-----	-----	-----	-----	-----
52802.652	52802.652	20148.303	0.3103498	0.3103498	1.178E-06	1.178E-06	7.359E-14

STIFFNESS MATRIX

MATRIX A				MATRIX B			MATRIX D		
-----				-----			-----		
168279.875	52225.625	0.000	0.000	0.000	0.000	122665.992	41864.426	4543.850	
52225.625	168279.891	0.000	0.000	0.000	0.000	41864.426	98432.148	4543.848	
0.000	0.000	58027.125	0.000	0.000	0.000	4543.850	4543.848	45874.426	

LOAD CASES

=====

Caso: 1

```

Nx = 1.595E+03      Mx = 0.00      Temp= 0.00
Ny = 1.514E+03      My = 0.00
Nxy = 261.          Mxy = 0.00
Defx = 7.400E-03     Kpx = 0.00
Defy = 6.700E-03     Kpy = 0.00
Defxy = 4.500E-03    Kpxy = 0.00

```


LOAD CASE NUMBER 1
=====

MAXIMUM STRAIN CRITERION (MODIFIED) - PROGRESSIVE FAILURE

PLY	AD/L-T	AD/L-C	AD/T-T	AD/T-C	AD/CORT	DEF-1	DEF-2	DEF-12	R.F.-1	R.F.-2	R.F.-12
----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----
1	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800-0.000700		1.3090	1.2875	3.9711
2	0.012174	0.010245	0.006180	0.005000	0.018085	0.004800	0.009300 0.000700		2.5363	0.6645	3.9711
3	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700 0.004500		1.6451	0.9224	3.9711
4	0.012174	0.010245	0.006180	0.005000	0.018085	0.006700	0.007400-0.004500		1.8170	0.8351	3.9711
5	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700 0.004500		1.6451	0.9224	3.9711
6	0.012174	0.010245	0.006180	0.005000	0.018085	0.006700	0.007400-0.004500		1.8170	0.8351	3.9711
7	0.012174	0.010245	0.006180	0.005000	0.018085	0.004800	0.009300 0.000700		2.5363	0.6645	3.9711
8	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800-0.000700		1.3090	1.2875	3.9711
9	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800-0.000700		1.3090	1.2875	3.9711
10	0.012174	0.010245	0.006180	0.005000	0.018085	0.004800	0.009300 0.000700		2.5363	0.6645	3.9711
11	0.012174	0.010245	0.006180	0.005000	0.018085	0.006700	0.007400-0.004500		1.8170	0.8351	3.9711
12	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700 0.004500		1.6451	0.9224	3.9711
13	0.012174	0.010245	0.006180	0.005000	0.018085	0.006700	0.007400-0.004500		1.8170	0.8351	3.9711
14	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700 0.004500		1.6451	0.9224	3.9711
15	0.012174	0.010245	0.006180	0.005000	0.018085	0.004800	0.009300 0.000700		2.5363	0.6645	3.9711
16	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800-0.000700		1.3090	1.2875	3.9711

ITERATION NUMBER: 2

DATA ENTRY

TITULO: EJEMPLO 4

PROPERTIES OF EACH MATERIAL

MAT	E11	E22	G12	N12	ALFA1	ALFA2
1	138000.000	8900.000	4700.000	0.3000000	-1.300E-06	3.130E-05
9	138000.000	0.000	0.000	0.0000000	-1.300E-06	0.000E+00

LAY-UP COMPOSITION

PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS
1	1	45.00	0.180	2	9	-45.00	0.180	3	1	0.00	0.180	4	1	90.00	0.180
5	1	0.00	0.180	6	1	90.00	0.180	7	9	-45.00	0.180	8	1	45.00	0.180
9	1	45.00	0.180	10	9	-45.00	0.180	11	1	90.00	0.180	12	1	0.00	0.180
13	1	90.00	0.180	14	1	0.00	0.180	15	9	-45.00	0.180	16	1	45.00	0.180

MEAN VALUES PROPERTIES OF THE LAY-UP

Exx	Eyy	Gxy	Nxy	Nyx	Alfax	Alfay	Alfaxy
50315.523	50315.527	19867.207	0.3259557	0.3259557	5.971E-07	5.971E-07	-1.188E-06

STIFFNESS MATRIX

MATRIX A				MATRIX B			MATRIX D		
162172.688	52886.438	-1466.329	0.000	0.000	0.000	118246.836	42342.586	3482.814	
52886.438	162172.703	-1466.333	0.000	0.000	0.000	42342.586	94012.992	3482.812	
-1466.329	-1466.333	57237.562	0.000	0.000	0.000	3482.814	3482.812	45303.098	

LOAD CASES

=====

Caso: 1

Nx = 1.548E+03	Mx = 0.00	Temp= 0.00
Ny = 1.471E+03	My = 0.00	
Nxy = 237.	Mxy = 0.00	
Defx = 7.400E-03	Kpx = 0.00	
Defy = 6.700E-03	Kpy = 0.00	
Defxy = 4.500E-03	Kpxy = 0.00	

LOAD CASE NUMBER 1

=====

MAXIMUM STRAIN CRITERION (MODIFIED) - PROGRESIVE FAILURE

PLY	AD/L-T	AD/L-C	AD/T-T	AD/T-C	AD/CORT	DEF-1	DEF-2	DEF-12	R.F.-1	R.F.-2	R.F.-12
1	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
2	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
3	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700	0.004500	1.6451	0.9224	3.9711
4	0.012174	0.010245	0.006180	0.005000	0.018085	0.006700	0.007400	0.004500	1.8170	0.8351	3.9711
5	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700	0.004500	1.6451	0.9224	3.9711
6	0.012174	0.010245	0.006180	0.005000	0.018085	0.006700	0.007400	0.004500	1.8170	0.8351	3.9711
7	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
8	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
9	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
10	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
11	0.012174	0.010245	0.006180	0.005000	0.018085	0.006700	0.007400	0.004500	1.8170	0.8351	3.9711
12	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700	0.004500	1.6451	0.9224	3.9711
13	0.012174	0.010245	0.006180	0.005000	0.018085	0.006700	0.007400	0.004500	1.8170	0.8351	3.9711
14	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700	0.004500	1.6451	0.9224	3.9711
15	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
16	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711

ITERATION NUMBER: 3

DATA ENTRY

TITULO: EJEMPLO 4

PROPERTIES OF EACH MATERIAL

MAT	E11	E22	G12	N12	ALFA1	ALFA2
1	138000.000	8900.000	4700.000	0.3000000	-1.300E-06	3.130E-05
9	138000.000	0.000	0.000	0.0000000	-1.300E-06	0.000E+00

LAY-UP COMPOSITION

PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS
1	1	45.00	0.180	2	9	-45.00	0.180	3	1	0.00	0.180	4	9	90.00	0.180
5	1	0.00	0.180	6	9	90.00	0.180	7	9	-45.00	0.180	8	1	45.00	0.180
9	1	45.00	0.180	10	9	-45.00	0.180	11	9	90.00	0.180	12	1	0.00	0.180
13	9	90.00	0.180	14	1	0.00	0.180	15	9	-45.00	0.180	16	1	45.00	0.180

MEAN VALUES PROPERTIES OF THE LAY-UP

Exx	Eyy	Gxy	Nxy	Nyx	Alfax	Alfay	Alfaxy
48486.898	50313.586	18692.023	0.3151473	0.3270200	-6.628E-07	6.328E-07	-1.296E-06

STIFFNESS MATRIX

MATRIX A				MATRIX B			MATRIX D		
155727.281	50952.812	-1466.329	0.000	0.000	0.000	115462.406	41507.262	3482.814	
50952.812	161592.609	-1466.334	0.000	0.000	0.000	41507.262	93762.383	3482.812	
-1466.329	-1466.334	53853.562	0.000	0.000	0.000	3482.814	3482.812	43841.215	

LOAD CASES

=====

Caso: 1

Nx = 1.487E+03	Mx = 0.00	Temp= 0.00
Ny = 1.453E+03	My = 0.00	
Nxy = 222.	Mxy = 0.00	
Defx = 7.400E-03	Kpx = 0.00	
Defy = 6.700E-03	Kpy = 0.00	
Defxy = 4.500E-03	Kpxy = 0.00	

LOAD CASE NUMBER 1

=====

MAXIMUM STRAIN CRITERION (MODIFIED) - PROGRESIVE FAILURE

PLY	AD/L-T	AD/L-C	AD/T-T	AD/T-C	AD/CORT	DEF-1	DEF-2	DEF-12	R.F.-1	R.F.-2	R.F.-12
1	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
2	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
3	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700	0.004500	1.6451	0.9224	3.9711
4	0.012174	0.010245	0.000000	0.000000	0.000000	0.006700	0.007400	0.004500	1.8170	0.0000	0.0000
5	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700	0.004500	1.6451	0.9224	3.9711
6	0.012174	0.010245	0.000000	0.000000	0.000000	0.006700	0.007400	0.004500	1.8170	0.0000	0.0000
7	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
8	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
9	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
10	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
11	0.012174	0.010245	0.000000	0.000000	0.000000	0.006700	0.007400	0.004500	1.8170	0.0000	0.0000
12	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700	0.004500	1.6451	0.9224	3.9711
13	0.012174	0.010245	0.000000	0.000000	0.000000	0.006700	0.007400	0.004500	1.8170	0.0000	0.0000
14	0.012174	0.010245	0.006180	0.005000	0.018085	0.007400	0.006700	0.004500	1.6451	0.9224	3.9711
15	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
16	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711

ITERATION NUMBER: 4

DATA ENTRY

TITULO: EJEMPLO 4

PROPERTIES OF EACH MATERIAL

MAT	E11	E22	G12	N12	ALFA1	ALFA2
1	138000.000	8900.000	4700.000	0.3000000	-1.300E-06	3.130E-05
9	138000.000	0.000	0.000	0.0000000	-1.300E-06	0.000E+00

LAY-UP COMPOSITION

PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS	PLY	MAT	ANGLE	THICKNESS
1	1	45.00	0.180	2	9	-45.00	0.180	3	9	0.00	0.180	4	9	90.00	0.180
5	9	0.00	0.180	6	9	90.00	0.180	7	9	-45.00	0.180	8	1	45.00	0.180
9	1	45.00	0.180	10	9	-45.00	0.180	11	9	90.00	0.180	12	9	0.00	0.180
13	9	90.00	0.180	14	9	0.00	0.180	15	9	-45.00	0.180	16	1	45.00	0.180

MEAN VALUES PROPERTIES OF THE LAY-UP

Exx	Eyy	Gxy	Nxy	Nyx	Alfax	Alfay	Alfaxy
48485.926	48485.926	17516.836	0.3157649	0.3157649	-6.412E-07	-6.412E-07	-1.419E-06

STIFFNESS MATRIX

MATRIX A				MATRIX B			MATRIX D		
155147.188	49019.188	-1466.329	0.000	0.000	0.000	115061.445	40170.738	3482.814	
49019.188	155147.188	-1466.334	0.000	0.000	0.000	40170.738	89307.312	3482.812	
-1466.329	-1466.334	50469.562	0.000	0.000	0.000	3482.814	3482.812	41502.184	

LOAD CASES

Caso: 1

Nx = 1.470E+03	Mx = 0.00	Temp= 0.00
Ny = 1.396E+03	My = 0.00	
Nxy = 206.	Mxy = 0.00	
Defx = 7.400E-03	Kpx = 0.00	
Defy = 6.700E-03	Kpy = 0.00	
Defxy = 4.500E-03	Kpxy = 0.00	

LOAD CASE NUMBER 1

MAXIMUM STRAIN CRITERION (MODIFIED) - PROGRESIVE FAILURE

PLY	AD/L-T	AD/L-C	AD/T-T	AD/T-C	AD/CORT	DEF-1	DEF-2	DEF-12	R.F.-1	R.F.-2	R.F.-12
1	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
2	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
3	0.012174	0.010245	0.000000	0.000000	0.000000	0.007400	0.006700	0.004500	1.6451	0.0000	0.0000
4	0.012174	0.010245	0.000000	0.000000	0.000000	0.006700	0.007400	0.004500	1.8170	0.0000	0.0000
5	0.012174	0.010245	0.000000	0.000000	0.000000	0.007400	0.006700	0.004500	1.6451	0.0000	0.0000
6	0.012174	0.010245	0.000000	0.000000	0.000000	0.006700	0.007400	0.004500	1.8170	0.0000	0.0000
7	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
8	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
9	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711
10	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
11	0.012174	0.010245	0.000000	0.000000	0.000000	0.006700	0.007400	0.004500	1.8170	0.0000	0.0000
12	0.012174	0.010245	0.000000	0.000000	0.000000	0.007400	0.006700	0.004500	1.6451	0.0000	0.0000
13	0.012174	0.010245	0.000000	0.000000	0.000000	0.006700	0.007400	0.004500	1.8170	0.0000	0.0000
14	0.012174	0.010245	0.000000	0.000000	0.000000	0.007400	0.006700	0.004500	1.6451	0.0000	0.0000
15	0.012174	0.010245	0.000000	0.000000	0.000000	0.004800	0.009300	0.000700	2.5363	0.0000	0.0000
16	0.012174	0.010245	0.006180	0.005000	0.018085	0.009300	0.004800	0.000700	1.3090	1.2875	3.9711

* THE MAXIMUM NUMBER OF ITERATIONS HAS BEEN EXCEEDED *
